

# How can we understand the usability of our tech?

- How do we identify problems?
- How do we pinpoint them to different aspects of usability?
- Can we quantify usability?

# 1. Heuristic Evaluation

- Find a checklist of required characteristics: does it do this?
- Check this with multiple reviewers with relevant expertise
- Include external reviewers to avoid dev-team bias

A nice quick starting point BUT

- Relatively shallow – focus on static qualities
- Cant deal with complex, issues that emerge in real contexts and interactions.
- You still didn't actually talk to the users!

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## Visibility of System Status

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.

- Does the design clearly communicate its state?
- Is feedback presented quickly after user actions?

Issues

Recommendations

3

## User Control and Freedom

Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process.

- Does the design allow users to go back a step in the process?
- Are exit links easily discoverable?
- Can users easily cancel an action?
- Is *Undo* and *Redo* supported?

Issues

Recommendations

<https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/>

Evaluation workbooks [here](#)

## 2. Task-based Evaluation (with screen recording)

- Choose a particular (relevant!) task your system should achieve
- Ask the user to try to achieve this and watch (carefully!) and take notes (in detail!)
- Must cover an “ecologically” wide range of activities (everyday but also exceptional “critical incidents”)

Like code-coverage in testing, task-covering is important here.

This requires a comprehensive list of use-cases, goals, and task-flows

# What are we looking for?

## How should we capture them?

- Quantitative aspects:

- Time
- Error rate
- Time-stamped action log
- “Human factors” questionnaires:
  - Cognitive load
  - Frustration

### Requirements:

- you must already expect it
- It must be ... quantifiable
- It must be readily captured without too much intrusion

- Qualitative aspects:

- Kinds of frustration
- Accessibility issues
- Qualities of experience
- Improvisatory behaviours

### Tools:

Free-text questionnaires

“talk aloud”

Interviews

Focus groups

Requires more skill and care to do well

More time-consuming

More informative

Captures unexpected issues

# Quant Questionnaires

Seem easy. Actually quite hard to do well

- Allows you to quantify “human factors” and make statistical comparisons
- Designing quantitative questionnaires is a science
  - Do your questions bias users?
  - Handling measurement error (measurement theory!)
  - Validating new questionnaires requires sophisticated statistical techniques and large numbers of users.
- Psychologists have made and validated some scales for you to use (like SUS, NASA TLX)
  - But do they quantify the things you need to know? Are the scores meaningful for your product?

## Likert (SUS)

Q1. I think I would like to use this system frequently.

☐ Strongly disagree   ☐ Disagree   ☐ Neither agree nor disagree   ☐ Agree   ☐ Strongly agree

Q2. I found the system unnecessarily complex.

☐ Strongly disagree   ☐ Disagree   ☐ Neither agree nor disagree   ☐ Agree   ☐ Strongly agree

Q3. I thought the system was easy to use.

☐ Strongly disagree   ☐ Disagree   ☐ Neither agree nor disagree   ☐ Agree   ☐ Strongly agree

# Qual Questionnaires

Also seem easy. Also actually quite hard to do well

- Bias is still a risk – leading questions
- How are you going to analyse it?
- Short questions and responses can be superficial
- More in-depth questions and responses require careful and rigorous analysis.

**Please tell us three things we can do to improve our service:**

In just one phrase, how would you describe Qualtrics culture?

After interacting with Qualtrics, what suggestions can you give us to help create an engaging culture?

# Structured Interviews

... harder than they look

- Similar to qualitative questionnaires but can be more responsive
- More in depth responses from users
- Requires a script of (semi) structured questions to work well
- Open-ended interviews are possible but require considerable skill, both in interview situation and during the analysis.
- Decide ahead of time how to analyse.

# Focus Groups

... you get the point by now

- Like interviews but in a small group, and you guide, or facilitate discussion, rather than ask questions from a script
- Make sure everyone gets to speak
- The larger group encourages some people to speak more freely and bounce ideas off each other
- But individual perspectives can be lost
- Requires a confident and adaptive facilitator

# Speak Aloud (with screen recording)

- Get the user to speak aloud as they perform a task, or use the technology freely (good to have both)
  - What they are thinking
  - What is hard to understand
  - Problems they encounter
  - Things that work well
- Advantages:
  - Doesn't rely on memory
  - Easy to connect problems to specific aspects of the software
    - Can make it easier to analyse
  - Uncovers issues we can't anticipate
- Disadvantages
  - Can prompt performance anxiety for users
  - Unreflective answers
  - Talking aloud can make behaviour unrealistic